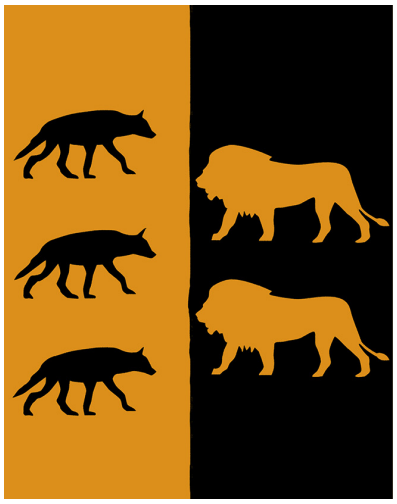


COMPLETE COMPETITORS CANNOT COEXIST

COMPETITIVE EXCLUSION PRINCIPLE

IN CONTEXT

KEY FIGURE

Georgy Gause (1910–86)

BEFORE

1925 Alfred James Lotka first uses equations to analyze variations in predator–prey populations, as does mathematician Vito Volterra, independently, a year later.

1927 Volterra enlarges and updates his 1926 study to include various ecological interactions within communities.

AFTER

1959 G. Evelyn Hutchinson extends Gause's ideas and produces a ratio describing the limit of similarity between two competing species.

1967 Robert MacArthur and Richard Levins use probability theory and Lotka–Volterra equations to describe how coexisting species interact.

Competition is the driver of evolution; the need to be bigger, stronger, and better inevitably leads to adaptations that give a species an edge. When two species compete for identical resources, the one which has any advantage will outdo the other. As a result, the weaker of the two species either becomes extinct or adapts, so that it no longer competes. This proposition,